

Optimized Topic Modeling of Nepal-Affiliated AI Research: Comparing Search Strategies and Tracking Themes, 2015–2025

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Abstract—Mapping a regional research literature requires models whose topics are both measurable and meaningful. We analyse 1,074 scholarly records retrieved as Nepal-affiliated AI research from 2015–2025, comparing grid-selected latent Dirichlet allocation (LDA), genetic-algorithm-optimised LDA, a budget-matched random-search control, and BERTopic. GA-LDA improves coherence over the standard baseline (C_v : 0.4483 to 0.4940), but random search finds a better solution in the single matched-budget experiment (fitness 0.5282 versus 0.5196). BERTopic achieves the highest C_v (0.5107) and three-seed topic stability (0.8418), while leaving 17.1% of documents unassigned. Three blinded LLM judgments favour BERTopic and rate GA-LDA below standard LDA, showing that improved coherence need not coincide with improved automated interpretability. An additional label audit identifies several GA topics as generic rather than domain-specific. Fixed-model temporal analysis shows increasing education-related topic share after 2020 under both representations. However, unequal window sizes, differing assignment coverage and a residual non-AI record limit interpretation, particularly for the eight-record earliest window. The results support reporting multiple quality measures, explicit coverage and qualitative checks rather than selecting a model from coherence alone.

Index Terms—topic modelling, latent Dirichlet allocation, genetic algorithms, BERTopic, Nepal, scientometrics

I. INTRODUCTION

A growing publication list does not by itself explain what a research community studies. Researchers choosing a project, institutions planning research support and readers entering a field need to distinguish recurring applications from emerging themes. Reading individual papers remains essential, but an overview becomes difficult when relevant work is scattered across engineering, health, education, agriculture and other disciplinary venues. Topic modelling offers a complementary way to organise this literature: it identifies patterns in document text that can be inspected, labelled and compared across publication periods.

This study examines research with at least one Nepal-based institutional affiliation. That definition concerns the affiliation recorded on a publication, rather than an author’s nationality or ethnicity. It includes internationally co-authored work when a Nepal affiliation is present and excludes diaspora-only work without one. The distinction matters because “research in Nepal” can otherwise refer to authorship, study location, language or national identity. Our evidence supports a view of indexed Nepal-affiliated literature, not a complete census of all AI activity in the country.

Latent Dirichlet allocation (LDA) [1] represents documents as mixtures of topics and topics as distributions over words. Its use in scientific literature analysis [2] provides an interpretable starting point for this application. Large-scale AI mapping

already includes temporal and country-level comparisons [3]; the question here is how useful different topic models are within a smaller regional corpus with heterogeneous applications and uneven temporal coverage. We investigate this particular setting without claiming that Nepal has never been studied or that regional analysis is itself novel.

Three methodological difficulties motivate the comparison. First, topic count and prior settings can change the granularity of an LDA summary: a broad topic may score consistently while failing to name a recognisable research area. Second, an embedding-based model may separate specific domains while declining to assign atypical documents, complicating comparisons with a model that represents every record. Third, publication growth and database coverage can be mistaken for thematic change unless topic prevalence is normalised and sample sizes remain visible. These are practical model-selection questions, not simply a contest to maximise one score.

We therefore compare a conventional LDA baseline, an evolutionary hyperparameter search, a random-search control and BERTopic on the same frozen corpus. The study contributes a documented regional dataset, an explicit test of whether the evolutionary mechanism adds value beyond search effort, and a temporal interpretation qualified by assignment coverage and topic quality. The genetic algorithm is an application of established methods rather than a new optimisation algorithm. The paper first positions the comparison in related work, then describes the corpus and methods, before examining quantitative results alongside the words and documents that give those results meaning.

II. RELATED WORK

A. Mapping scientific literature

Griffiths and Steyvers [2] demonstrate how probabilistic topics can organise scientific publications. Yu and Xiang [3] apply LDA to 177,204 AI documents from 1990–2021, extracting 40 topics and aggregating them by year, publication source and country or region. Their work establishes that topic prevalence can provide a structured account of AI research. Our contribution differs in scale and emphasis: a Nepal-affiliated corpus is analysed through competing representations and explicit evaluation of their limitations. A global model’s topic vocabulary cannot be assumed to offer the most useful resolution for this narrower collection.

B. Search and embedding-based topic models

Agrawal, Fu and Menzies [4] investigate instability in topic modelling and use search-based optimisation to improve model configurations. This motivates searching LDA’s

hyperparameters instead of fixing them arbitrarily. However, improvement over a small grid does not isolate the contribution of an evolutionary algorithm. Random search is an established hyperparameter-selection baseline [5]; matching its space, objective and number of unique evaluations to the GA makes the comparison more informative.

BERTopic [6] combines document embeddings with clustering and class-based TF-IDF topic representations. Its components include Sentence-BERT [7], UMAP [8] and HDBSCAN [9]. Capilitan et al. [10] illustrate its application to a large agricultural AI literature, showing the relevance of embedding-based thematic mapping to an applied domain. Its usefulness on our collection is an empirical question: density clustering can discover specific groups but may leave some records outside those groups. We use hard assignments for temporal reporting; this is a choice in our implementation, not a claim that BERTopic cannot produce probability estimates.

C. Evaluation beyond a single score

Chang et al. [11] show why statistical fit and human interpretation should not be treated as equivalent. Word co-occurrence approaches [12], [13] and the C_v framework of Röder et al. [14] provide measurable aspects of topic quality, while topic diversity [15] addresses repeated vocabulary across topics. None alone establishes that a topic names a useful research domain. We therefore combine coherence, diversity, stability, document coverage and blinded automated judgments. The latter are explicitly provisional: this experiment does not replicate a human interpretation study. The combined evaluation asks which compromises each model makes and whether those compromises suit a regional literature review.

III. PROBLEM AND DATASET

A. Research questions

RQ1: Which themes are identifiable in the retrieved Nepal-affiliated literature? **RQ2:** How does their prevalence change between publication windows? **RQ3:** Does GA-LDA improve on grid-selected LDA and on budget-matched random search? **RQ4:** How do these configurations compare with BERTopic in quality and coverage?

B. Retrieval and inclusion

Records were retrieved from OpenAlex [16] using a Nepal-affiliation country filter, AI-related topic subfields and a supplementary keyword pass for applied work indexed elsewhere. The inclusion rule required a Nepal-based institutional affiliation and AI-related vocabulary in the title or abstract. The analysis uses concatenated titles and abstracts rather than full texts. The filtering sequence is reported in Table I.

OpenAlex’s structured affiliation metadata made it suitable for the geographical inclusion rule. This departs from the proposal’s planned Semantic Scholar-led, multi-source collection: NepJOL, IEEE and Springer were not separately harvested into this analysis. Indexing through OpenAlex may nevertheless include records from those publishers or outlets. The absence of direct supplementation limits coverage; it is not proof

TABLE I
FROZEN CORPUS INCLUSION SEQUENCE.

Stage	Records remaining
OpenAlex retrieval	1,693
Publication year through 2025	1,228
Abstract available	1,102
At least 30 title/abstract words	1,088
Exclude computer vision syndrome	1,074

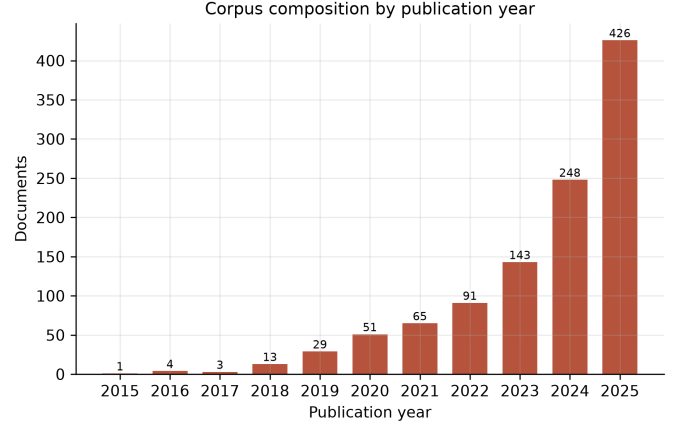


Fig. 1. Publication counts in the frozen corpus. The last window contains 674 of 1,074 records; growth reflects both research activity and retrieval coverage.

that every local publication is absent, nor that supplementary harvesting is technically impossible.

The corpus contains 560 distinct nonempty venue names and a median title/abstract length of 212 words. Venue metadata classifies 1,004 records as published, 36 as preprint-server records and 34 as lacking a venue. Thus, 93.5% have publication-venue evidence, but the dataset is not a verified wholly peer-reviewed collection as originally proposed. Recorded affiliation strings are available for 1,072 records. An earlier 30-record affiliation check reported no incorrect Nepal affiliations; that limited check does not establish complete AI relevance or coverage.

C. Quality checks and temporal coverage

The four windows contain 8, 93, 299 and 674 records respectively (Fig. 1). These counts make the earliest window unsuitable for a separately fitted topic model. Fourteen “computer vision syndrome” papers were removed because their use of that phrase concerned screen-related eye conditions rather than computer vision methods. This illustrates why keyword matching requires inspection of context.

A subsequent document check found another ambiguity: a 2016 medical-education paper discusses students’ “deep learning” approaches, not machine learning. It remains in the archived training corpus, so the comparison is reported as a study of the frozen retrieval rather than a perfectly adjudicated AI dataset. We record its identifier and quantify its influence on the earliest window in Section VI-C. Policy and ethics papers about AI are also retained. These qualifications prevent

apparent thematic precision from concealing uncertainty in inclusion.

Affiliation validity and subject relevance are separate questions. An institution can be correctly identified while the record is included for the wrong sense of a technical phrase. A useful adjudication process therefore checks the abstract’s methods as well as title keywords, records the reason for each disputed inclusion, and preserves identifiers for reproducible exclusions. Here the residual error is disclosed rather than removed only from the model with an unfavorable score. Revising the training corpus would require rerunning the entire comparison, since even one document can change vocabulary frequencies, cluster membership or selected hyperparameters.

IV. METHODS

Figure 2 summarises the shared corpus and model-specific representations.

A. Text preparation and feature extraction

Cleaning normalises Unicode, removes leading abstract markers, URLs and DOI strings, and consolidates parenthesised acronym glosses. LDA’s branch lowercases and tokenises the text, removes stop words, applies noun/verb lemmatisation and detects bigram/trigram phrases. Technical acronyms such as AI, ML, NLP and CNN are protected from the minimum-length filter and lemmatiser. The dictionary retains terms appearing in at least five documents and at most half the corpus, yielding 3,018 terms and no empty documents. LDA receives integer bag-of-words counts; the pipeline also exports TF-IDF features, but these are not used to train the probabilistic model.

The domain stop-word list removes publication boilerplate and frequent generic research-process terms. Its development used measured document frequency alongside semantic judgment; frequency alone cannot distinguish a generic term from an important subject. For example, “level” and “value” were retained because they can carry hydrological or statistical meaning. Terms such as “data”, “accuracy” and “performance” also remain. Their persistence in some topics is therefore a limitation of the representation, not evidence that cleaning removed them successfully.

BERTopic receives cleaned natural sentences for embedding. Removing function words and word order before sentence encoding would discard information the encoder expects. The topic-word representation uses the configured stop words, but tokenisation and phrase vocabularies still differ between branches. Shared documents and a shared stop list do not make preprocessing identical; this difference is considered when interpreting coherence.

B. LDA baseline and genetic search

LDA is fitted with Gensim [17] using variational inference [18]. The baseline selects K from $\{5, 10, \dots, 40\}$ by maximum C_v , with symmetric document priors and the library’s default topic-word prior. The selected value is $K = 10$ (Fig. 3); it is a searched baseline, not an arbitrary topic count.

The genetic algorithm [19] searches (K, α, η) , where Gensim’s η is the topic-word Dirichlet prior often denoted β . Topic count ranges from 5 to 40, and each scalar prior ranges from 0.01 to 1.0. Initial priors are sampled log-uniformly to cover sparse and dense settings. Chromosomes use an integer topic count and real-valued priors. Tournament selection, blend crossover, Gaussian mutation and two-member elitism generate subsequent populations. Mutation is scaled to the gene range; bounds and valid integer topic counts are enforced after variation.

Fitness scalarises three objectives:

$$F = 0.6\widehat{C}_v + 0.2D_{25} + 0.2S_{10},$$

$$\widehat{C}_v = \text{clip}\left(\frac{C_v - 0.30}{0.40}, 0, 1\right). \quad (1)$$

Here D_{25} is the fraction of unique words across top-25 lists, and S_{10} is matched top-10-word stability. The weights express a preference for coherence; they are not learned and no Pareto frontier is estimated. Alternative weights could select different configurations. The historical GA evaluates 20 individuals over 15 generations; caching duplicate genomes reduces 300 candidate requests to 183 unique evaluations, each requiring three LDA fits.

C. Random search and BERTopic

Random search uses the same hyperparameter bounds, log-uniform initial sampling rule, fitness, LDA settings and stability seeds. Its budget is 183 unique configurations, matched to the GA’s actual evaluations. This controls for searching a larger space than the eight-point baseline. The strategies differ in how subsequent candidates are proposed; matching evaluations does not imply identical wall-clock overhead or equal performance over repeated search runs.

BERTopic uses `all-MiniLM-L6-v2` embeddings, UMAP with 15 neighbours and five output dimensions, and HDBSCAN with minimum cluster size 15 and excess-of-mass cluster selection. Its 14 non-outlier clusters provide the topics used here. The outlier label is retained in coverage calculations instead of being reassigned to improve apparent completeness.

D. Stability and temporal summaries

Across each pair of fitted seeds, topic word sets are matched one-to-one by maximising total Jaccard overlap with Hungarian assignment. The score averages matched overlap across topics and seed pairs, penalising unequal topic counts by their size ratio. This avoids treating arbitrary topic indices as meaningful. It measures reproducibility of top words, not agreement of document assignments or complete topic distributions.

One global model defines each topic vocabulary across all years. LDA’s temporal share is the mean document-topic mass within a window. BERTopic’s share is the fraction of assigned documents in a cluster; zero-mass outlier rows are excluded. Its denominator therefore differs from LDA’s. We show assigned and total counts for every window and exclude

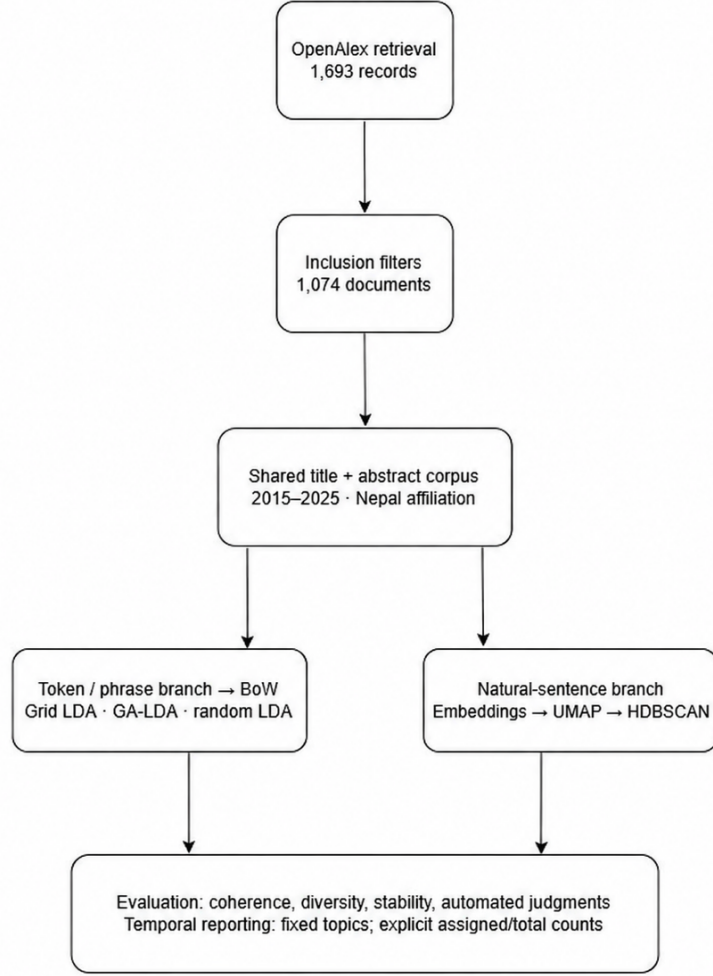


Fig. 2. End-to-end analysis. All models use the same records, while representations differ to suit the model. Label assessment and temporal interpretation follow, rather than define, training.

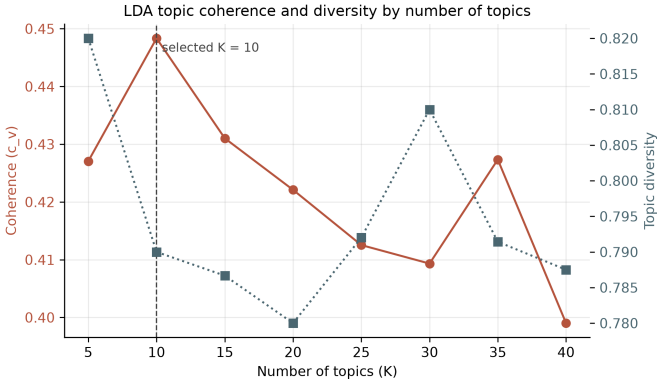


Fig. 3. Baseline topic-count sweep. Selection uses C_v only; the displayed diversity is $D@10$ and is a diagnostic rather than a selection criterion.

2015–2017 from trend fitting. This replaces the proposal’s per-window refitting; prevalence changes are measured under fixed topics, without claiming to identify topic births or changes in

topic meaning.

E. Automated thematic labelling

All 39 topics have archived LLM-assisted draft labels, followed by an automated label audit. The drafts were produced in an assistant session, not through provider API calls. This fulfils the proposal’s automated-labelling step; it does not establish human validation.

A Streamlit app exposes the process through a wrapper for OpenAI, Claude and Gemini. It previews top words and up to four representative titles, generates structured labels with rationale and evidence identifiers, and records prompts, source hashes, requested/resolved models, responses and usage. Random-search topics use words only because their archived run lacks document-topic assignments. Failed batches can resume; named human reviews and partial exports remain explicit. Provider adapters were tested with simulated responses; no new live API run was made because credentials

were unavailable. The reported results retain their archived labels and judgments.

V. EXPERIMENTAL SETUP

The archived search seed is 42, with seeds 42–44 used for stability. LDA uses ten passes, 100 iterations and batches of 500 documents. GA settings are population 20, 15 generations, tournament size three, crossover probability 0.7, mutation probability 0.05 and mutation scale 0.15 times the gene range. Coherence is evaluated in a single process. The comparison reports C_v , c_{npmi} and u_{mass} on top-10 words, and diversity/Jaccard on top-10 lists. **The reported D@10 differs from D@25 inside search fitness**; both cutoffs are explicit so the table is not mistakenly read as the fitness decomposition.

Perplexity is evaluated on the training corpus and is an in-sample variational diagnostic. Metrics not used in fitness provide additional views, but are not independent held-out tests. The original three-seed stability also reuses selection seeds. A separate fixed-configuration check on seeds 100–109 is reported as a sensitivity analysis, not as repeated optimisation or out-of-sample accuracy. No chronological prediction task is claimed.

For BERTopic coherence, words missing from the shared LDA dictionary are dropped; 93 of 140 topic words remain. The resulting scores therefore compare different effective word-list lengths, an important limitation of numerical ranking. A topic with fewer than two retained words would be omitted by the scorer. Embedding-based topic quality should not be inferred from the filtered score alone.

The interpretability instrument presents shuffled top-10 lists and a 1–5 rubric to three blinded LLM judgments. Model names, candidate labels and other judgments are withheld. A separate automated reviewer assesses all 39 candidate labels as accept, revise or reject. The records identify the judging model as `claude-opus-5`; separate contexts are recorded, but sampling settings are not fully preserved. Thus the aggregation is reproducible from saved ratings, whereas an identical fresh judgment cannot be guaranteed. Ordinal Krippendorff’s α [20] measures agreement. Shared model biases remain possible, and these ratings are not a substitute for human validation.

To assess sampling sensitivity of the education trend, we resample documents within the 2018–2020 and 2024–2025 windows 2,000 times and take percentile intervals for the difference in mean share. These intervals condition on the fitted model and, for BERTopic, assigned documents. They exclude uncertainty from retrieval, topic fitting, labelling and outlier selection.

VI. RESULTS

A. Model comparison and search behaviour

For RQ3, GA-LDA improves C_v by 10.2% relative to the grid baseline and also improves the other coherence measures, reported diversity, original-seed stability and in-sample perplexity (Table II). It does not improve every metric: standard LDA has less pairwise topic overlap and a higher automated interpretability score. Coverage is unchanged.

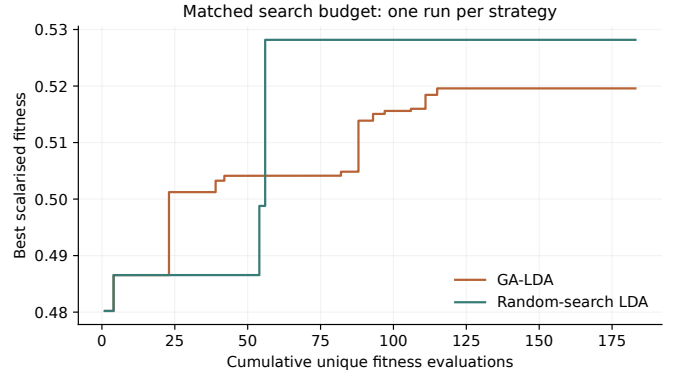


Fig. 4. Best-so-far search fitness against actual unique evaluations, reconstructed from the saved candidate histories. Each curve represents one search run; no uncertainty band is implied.

Random search finds the better scalarised fitness, 0.5282 versus 0.5196, with $K = 8$ rather than the GA’s $K = 7$. Figure 4 compares incumbent fitness at the same cumulative number of unique evaluations. The GA stops improving at generation eight, whereas the random run eventually finds one candidate exceeding the GA result. This supports a conclusion about the observed runs only. It does not establish that random search generally wins, that GA reduces variance, or that evolutionary search contributes nothing in other settings.

For RQ4, BERTopic leads in C_v and original-seed stability but has lower diversity and poorer c_{npmi} and u_{mass} . It leaves 184 documents unassigned. Its top words are comparatively reproducible across the three seeds, but filtered vocabulary and different representations qualify direct metric comparisons. No single row resolves the practical choice between complete document representation and more specific clusters.

On ten additional seeds (100–109), matched-word stability for the fixed standard LDA, GA-LDA and random-search configurations is 0.2718, 0.3525, 0.3581, respectively. The same ordering persists, but absolute LDA stability remains modest. These 45 pairwise comparisons per configuration share fitted runs and are not 45 independent experiments; no significance test between search strategies is inferred from them.

B. Interpreting the topics

The panel’s ordinal agreement is $\alpha = 0.9787$, with 93.2% pairwise exact agreement and all rating pairs within one point. The ordering is BERTopic, random search, standard LDA, then GA-LDA (Fig. 5). This is *not* a complete reversal of C_v : the first two models retain their ranks, while the two remaining LDA configurations exchange positions. The narrower finding is useful: GA-LDA’s coherence gain over standard LDA does not translate into a higher rating under this automated rubric.

Figure 6 shows why. GA topic 5 is dominated by evaluation and prediction vocabulary and cannot confidently be called a crop or hazard theme. Other clusters name more specific domains. The label audit accepts 20 labels, revises 12 and rejects seven (Fig. 7). Only one of seven GA labels is accepted unchanged, compared with nine of fourteen BERTopic labels.

TABLE II
ARCHIVED MODEL COMPARISON. ARROWS INDICATE PREFERRED METRIC DIRECTION, NOT UNIVERSAL MODEL UTILITY. STABILITY USES THE ORIGINAL THREE SEEDS; PERPLEXITY IS IN-SAMPLE.

Metric	Standard LDA	GA-LDA	Random-search LDA	BERTopic
Topics	10	7	8	14
$C_v \uparrow$	.4483	.4940	.5030	.5107
$c_{\text{npmi}} \uparrow$	-.0230	.0112	.0245	-.0300
$u_{\text{mass}} \uparrow$	-3.3586	-2.1796	-2.5477	-3.4444
Diversity D@10 $\uparrow$	.7900	.8000	.8000	.7286
Within-model Jaccard J@10 $\downarrow$	.0359	.0471	.0536	.0471
Stability S@10 $\uparrow$	.2526	.3375	.3382	.8418
In-sample perplexity $\downarrow$	172.1	167.8	165.9	n/a
Unassigned documents	0%	0%	0%	17.1%
Mean LLM rating (1–5)	2.57	2.05	2.88	4.10
Labels accepted unchanged	5/10	1/7	5/8	9/14

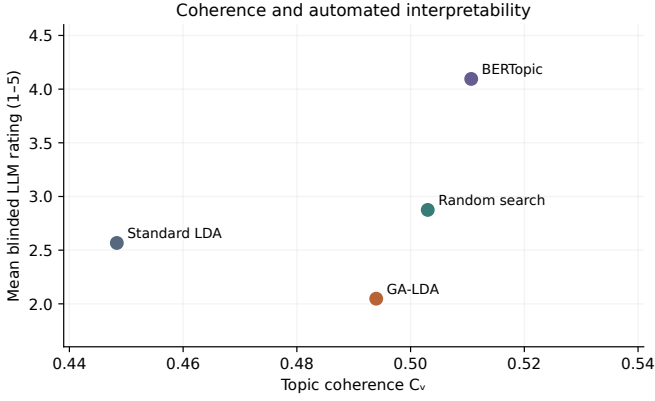


Fig. 5. Model-level coherence and automated interpretation scores. Four descriptive points do not establish a correlation or causal effect of optimisation.

This separate task corroborates a difficulty in assigning useful labels, although its judgments can share biases with the rating panel. Broad topics and low ratings are associated in this comparison; a controlled topic-count ablation would be needed to attribute the difference to K alone.

C. Themes and temporal prevalence

The audited labels identify applied domains including medical imaging, drug discovery, cybersecurity, construction materials, water/land modelling and Nepali text processing. These provide an answer to RQ1 at the resolution supported by the retrieved corpus. Mixed GA topics remain labelled as mixed; they are not recast as specific applications to make a cleaner narrative.

For RQ2, GA’s education-related share rises from 2.1% in 2018–2020 to 15.2% in 2024–2025, an increase of approximately 13.1 percentage points. BERTopic’s education cluster rises from 1.4% to 19.5% of assigned records, approximately 18.1 points. Conditional bootstrap intervals are 10.7–15.4 and 13.7–22.2 points respectively. Both fitted representations therefore indicate an education-related increase across these endpoints, although the values are not directly interchangeable and the intervals do not address shared coverage bias.

Figure 10 makes this denominator effect explicit. BERTopic assigns 62.5% of the earliest records, 77.4% in 2018–2020, 83.3% in 2021–2023 and 83.7% in 2024–2025. Its changing coverage could therefore influence a temporal comparison conditioned on assignment. When the education cluster is instead expressed as a fraction of *all* retrieved documents, including outliers in the denominator, its share rises from approximately 1.1% to 16.3% between the later endpoints. The increase persists under this alternative summary, although outliers still have unknown thematic membership. Counting them as not assigned to education does not establish that none is about education; it measures the observed cluster’s corpus share.

This check separates two possible explanations for the displayed rise: a greater fraction of papers being assigned somewhere, and a larger observed education cluster. It strengthens the descriptive finding without implying that the cluster is a complete or unbiased inventory of educational AI. Reading its representative documents and adjudicating unassigned records would be needed to assess that stronger claim. The same distinction should accompany any proposed comparison of thematic prevalence across model families.

The earliest GA education share requires particular caution. The non-AI medical-education record receives 0.721 education-topic mass. Excluding that record from reporting alone reduces the earliest share from about 10.8% to 2.1%, without retraining. This is a sensitivity check, not a corrected model fit. It demonstrates that a modern label applied retrospectively to a fixed topic cannot establish early generative-AI activity. The entire earliest window remains descriptive rather than evidence for a decade-long trajectory.

Declining relative shares for some image/text and mixed forecasting topics do not imply falling publication counts: the corpus grows sharply. Likewise, a broad education cluster cannot be equated wholly with generative AI. The defensible interpretation concerns topic prevalence in the retrieved literature, not a causal account of changes in national research priorities.

GA-LDA T5: generic evaluation

- 1 accuracy
- 2 data
- 3 performance
- 4 classification
- 5 prediction
- 6 image
- 7 detection
- 8 dataset
- 9 algorithm
- 10 technique

GA-LDA T4: education

- 1 ai
- 2 student
- 3 education
- 4 tool
- 5 learn
- 6 financial
- 7 generative ai
- 8 technology
- 9 skill
- 10 influence

BERTopic T11: drug discovery

- 1 drug
- 2 molecular
- 3 inhibitors
- 4 cancer
- 5 discovery
- 6 chemical
- 7 ai
- 8 compounds
- 9 screening
- 10 binding

Fig. 6. Actual top-10 lists illustrate what aggregate scores can hide: a generic evaluation topic, an education-related topic and a domain-specific drug-discovery cluster.

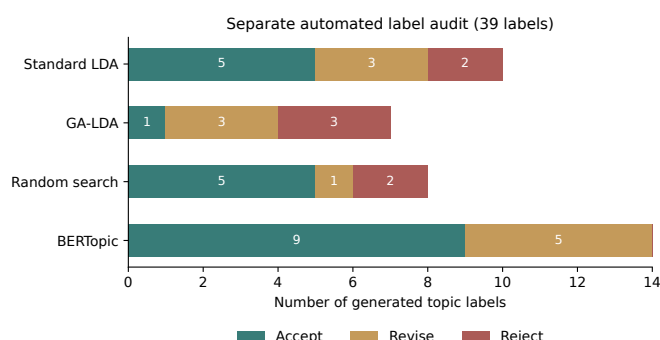


Fig. 7. Automated review outcomes, with counts shown rather than treating models with different numbers of topics as equally sized samples. Human label validation remains outstanding.

VII. SOCIAL, ETHICAL, LEGAL AND PROFESSIONAL CONSIDERATIONS

The study concerns an identifiable research community. Affiliation is used for inclusion, not to infer nationality or rank individual researchers. Findings are reported at topic level; neither publication counts nor topic scores are measures of a person’s research quality.

OpenAlex supplies openly available scholarly metadata. The collection uses metadata and abstracts rather than bypassing paywalls, and the scraper records source identifiers with rate-limited requests. Metadata availability should not be read as blanket permission to redistribute every underlying text; provenance and applicable source terms remain relevant. Direct supplementation of locally published and Nepali-language scholarship was not completed, which may underrepresent precisely the work of local interest.

LLMs assisted topic labelling, automated assessment and manuscript preparation. Their outputs are fallible: generated labels were revised or rejected, and the current report distinguishes automated judgments from human review. The authors remain responsible for the submission, its sources

and its interpretation. Claims of perfectly independent judges or guaranteed fresh reproduction of a judgment are avoided. Recorded ratings, prompts and review status support inspection of the process without representing it as a human study.

Professional reporting also requires retaining unfavorable results. The matched control and generic topics are therefore reported alongside positive coherence changes. Archived outputs are not silently overwritten when data-quality concerns arise; changes to inclusion or training would require a new versioned experiment and a renewed comparison.

VIII. DISCUSSION AND CONCLUSIONS

The comparison illustrates a distinction between improving a metric and improving a useful regional summary. Fine hyperparameter search improves several LDA diagnostics relative to the grid baseline, but the observed GA does not outperform the matched random-search run. Neither result establishes a universal ordering of optimisers. BERTopic provides more recognisable domain labels under the automated rubric, at the cost of incomplete document assignment and a coherence vocabulary that only partly overlaps the common reference dictionary.

For literature exploration, BERTopic’s specific clusters are useful starting points for reading. For summaries requiring a distribution for every record, LDA offers complete representation, but its mixed topics should retain cautious labels. The practical recommendation is to inspect topic words and representative documents alongside metrics and coverage, rather than allowing the highest score to determine the interpretation automatically.

The limitations are substantive. The frozen corpus contains a known residual relevance error, has sparse early coverage and does not exhaust local sources. Stability tests concern word lists on the same corpus, automated ratings are not human validation, and only one complete run of each search strategy is compared. Further work should adjudicate ambiguous records, rerun all models on the revised corpus, repeat

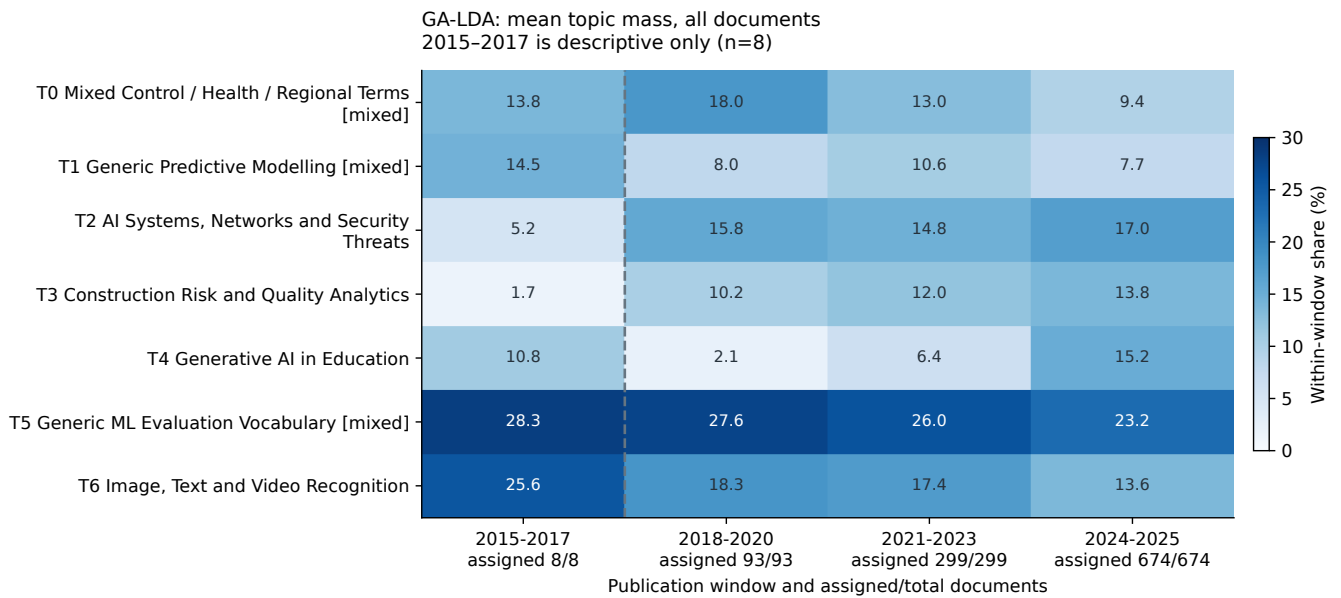


Fig. 8. GA-LDA topic mass by window, using audited labels and all seven topics. Values are percentages over all documents. The dashed boundary separates the eight-record descriptive window.

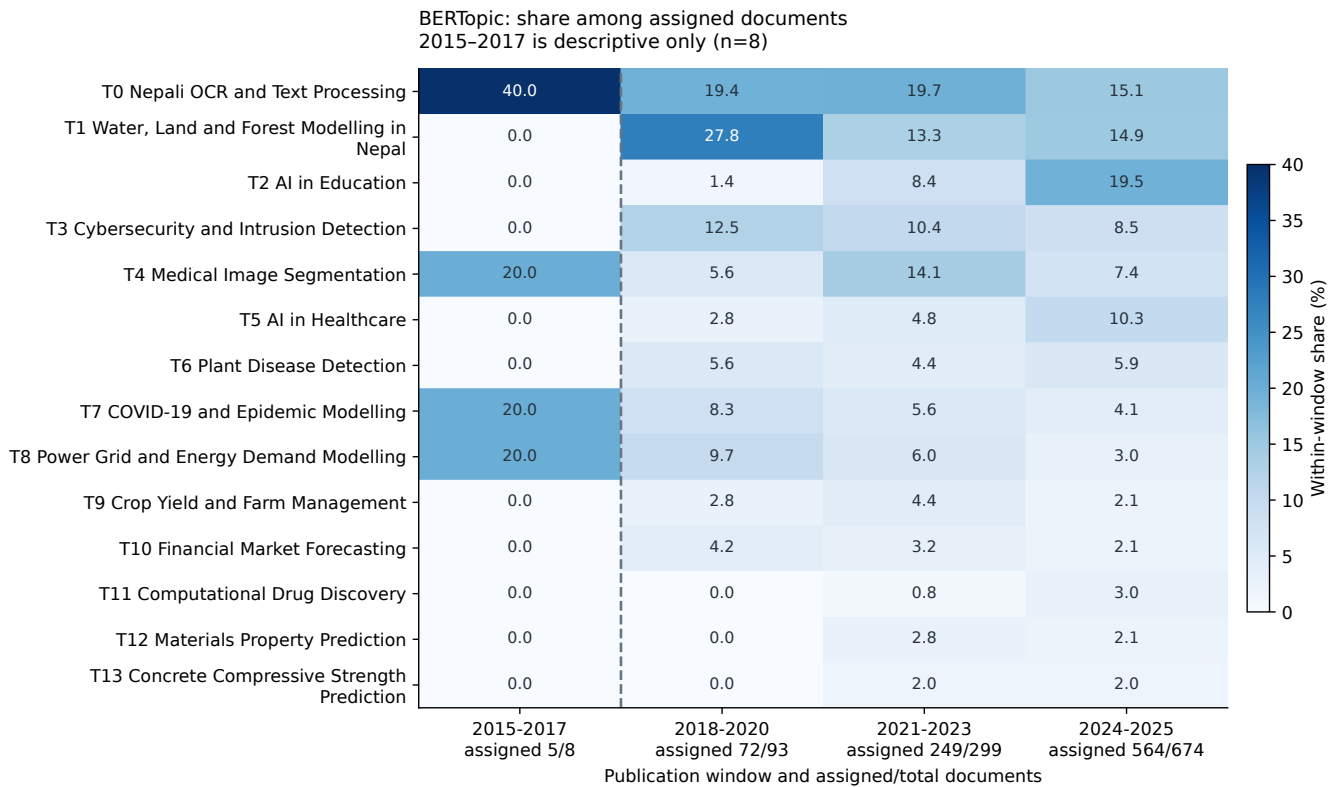


Fig. 9. BERTopic cluster prevalence conditional on assignment. Assigned counts are 5, 72, 249 and 564; total counts are 8, 93, 299 and 674. These denominators differ from Fig. 8.

matched searches, examine sensitivity to fitness weights and topic count, and obtain human judgments using the existing blinded instrument.

Within these limits, the study provides a reproducible com-

parison and evidence of increased education-related prevalence in the later windows. Its main methodological lesson is that coherence, interpretability, stability and coverage answer different questions. Making their definitions and disagreements

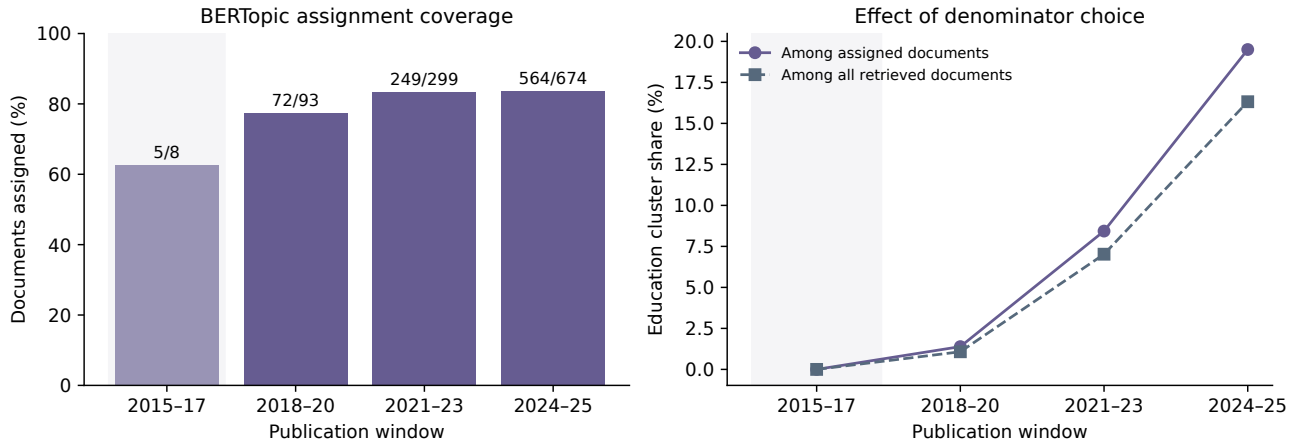


Fig. 10. Assignment coverage and a denominator sensitivity check for BERTopic’s education cluster. The grey region marks the eight-record early window. Counting outliers in the total reduces the estimated level, while the later-window increase persists.

visible produces a more credible account of regional research than selecting one model and treating its labels as established facts.

INDIVIDUAL CONTRIBUTIONS

Bibek Paudyal (250288) coordinated corpus retrieval, pre-processing and the topic-labelling application. Sajan Mahat (250289) coordinated the LDA baseline, genetic search and matched random-search experiments. Siddhartha Bhatta (250620) coordinated BERTopic, model evaluation and temporal analysis. All three members contributed equally to report preparation and review of the implementation, results and interpretation, with shared responsibility for the final submission.

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APPENDIX: CODE AND REPRODUCIBILITY

The implementation and experiment configuration are available at

<https://github.com/bpdyl/ai-research-topic-modeling>

Pinned runs, source hashes and figure statistics are recorded in `paper/report_evidence.json`. Notebook quick mode checks execution; full settings reproduce the protocol. Automated assessments use saved judgments.